SIMATS ENGINEERING
ASSESSMENT TOOL 30

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I HEISENBERG RAJA(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Scenario based Questions

Q1. Data Cleaning and Normalization in Healthcare Analytics (L3, L4, L5)

A healthcare analytics company is developing a model to predict patient readmission. The dataset collected from different hospitals contains missing values, inconsistent categorical entries, and extreme values.

Sample Dataset

Patient_ID	Age	Gender	BMI	Annual_Visits	Blood_Pressure	Risk_Level
P1	25	M	22.5	3	120	Low
P2	NaN	Male	24.8	5	130	Medium
P3	42	F	NaN	7	135	Medium
P4	110	Female	45.2	50	200	High
P5	36	female	29.5	NaN	140	Medium
P6	50	FEMALE	31.2	12	150	High
P7	29	male	21.8	4	125	Low
P8	45	Female	60.0	40	190	High

Tasks

A. Missing Value Treatment

i. Identify missing values in Age, BMI, and Annual_Visits.

ii. Apply Median Imputation and Mean Imputation.

iii. Compare the resulting values and justify which method is more appropriate.

B. Outlier Detection

i. Detect outliers in Age and Annual_Visits using the Z-score method.

ii. Identify the records containing extreme observations.

iii. Apply capping or removal and justify the selected approach.

C. Categorical Standardization

i. Identify inconsistent Gender values.

ii. Convert all values into Male/Female.

iii. Explain why categorical standardization is important for machine-learning models.

D. Normalization

i. Apply Min-Max normalization to BMI.

ii. Explain the effect of normalization on model performance.

Q2. Covariance and Feature Selection in Banking Analytics (L4, L5)

A bank wants to develop a customer-risk prediction model. Several attributes are strongly related, resulting in redundant information and increased computational complexity.

Sample Dataset

Customer_ID	Age	Income (₹K)	Loan_Amount (₹K)	Credit_Score	Monthly_EMI (₹)	Savings (₹K)	Risk
B1	25	35	100	650	5000	20	Low
B2	32	50	150	680	7000	30	Low
B3	40	70	200	720	9000	45	Medium
B4	48	100	300	780	14000	70	High
B5	29	42	120	660	5500	25	Low
B6	38	65	180	710	8500	40	Medium
B7	55	130	350	820	17000	90	High
B8	44	85	250	760	12000	60	High

Tasks

A. Covariance Analysis

i. Calculate the covariance between Income and Loan_Amount.

ii. Determine whether the relationship is positive or negative.

iii. Explain what a high positive covariance indicates in this dataset.

B. Feature Relationship Analysis

i. Identify pairs of attributes that may be highly correlated.

ii. Explain why Age, Income, Savings, and Loan_Amount may contain redundant information.

iii. Identify at least three features that could be removed.

C. Feature Selection

i. Select an optimal subset of features for predicting Risk.

ii. Justify your selection using correlation and domain knowledge.

iii. Explain how feature selection reduces computational cost.

D. Standardization

i. Standardize all numerical attributes using Z-score normalization.

ii. Explain why Credit_Score and Income cannot be directly compared without scaling.

Q3. Discretization of Sales Data in Retail Analytics (L3, L4, L5)

A retail company wants to classify customers based on the amount they spend per month. Continuous spending values are difficult for managers to interpret, so the company plans to use discretization.

Sample Dataset

Customer_ID	Monthly_Spend (₹)	Transactions	Items_Bought	Customer_Category
C1	500	4	6	Basic
C2	750	5	9	Basic
C3	1000	7	12	Basic
C4	1500	9	18	Silver
C5	2000	12	25	Silver
C6	2500	14	30	Gold
C7	3000	16	36	Gold
C8	4000	20	45	Gold
C9	5000	25	55	Platinum
C10	6500	30	70	Platinum

Tasks

A. Equal-Width Binning

i. Divide Monthly_Spend into 4 equal-width bins.

ii. Calculate the bin width.

iii. Define the intervals and assign each customer to a bin.

B. Equal-Frequency Binning

i. Divide the dataset into 4 bins containing approximately equal numbers of records.

ii. Assign each customer to the corresponding bin.

iii. Compare equal-width and equal-frequency results.

C. Concept Hierarchy

Construct a concept hierarchy:

Monthly_Spend → Low → Medium → High → Very High

D. Interpretation

i. Explain how discretization improves decision-making.

ii. Discuss one disadvantage of converting continuous values into categories.

Q4. Data Transformation in Manufacturing Quality Analytics (L3, L4, L5)

A manufacturing company records machine operating temperatures. The measurements vary considerably, and some machines produce unusually high readings.

Sample Dataset

Machine_ID	Temperature (°C)	Operating_Hours	Vibration
M1	55	100	2.1
M2	60	120	2.5
M3	62	140	2.8
M4	65	160	3.0
M5	70	180	3.2
M6	75	200	3.8
M7	80	220	4.1
M8	120	250	8.5

Tasks

A. Min-Max Normalization

i. Normalize Temperature into the range [0,1].

ii. Calculate the normalized value for each machine.

iii. Identify which observation obtains the maximum normalized value.

B. Z-Score Normalization

i. Calculate the mean Temperature.

ii. Calculate the standard deviation.

iii. Compute the Z-score for each temperature value.

C. Outlier Analysis

i. Identify whether 120°C can be considered an extreme value using the Z-score method.

ii. Explain how this value affects mean-based normalization.

iii. Suggest an appropriate treatment method.

D. Comparison

Compare Min-Max and Z-score normalization in terms of their sensitivity to extreme observations.

Q5. Missing Value Imputation and Data Transformation in Student Analytics (L3, L4, L5)

An educational institution is building a student-performance prediction model. Data collected from different departments contain missing examination marks, attendance values, and study hours.

Sample Dataset

Student_ID	Study_Hours	Internal_Mark	Attendance (%)	Assignment_Score	Grade
S1	2	55	70	60	C
S2	3	NaN	75	65	C
S3	4	68	NaN	72	B
S4	5	75	85	78	B
S5	NaN	80	90	82	A
S6	7	88	92	NaN	A
S7	8	92	95	90	A
S8	12	98	98	95	A

Tasks

A. Missing Value Analysis

i. Identify all missing observations.

ii. Apply mean imputation and median imputation.

iii. Determine which method is more suitable for Study_Hours and explain why.

B. Data Transformation

i. Apply Min-Max normalization to Internal_Mark.

ii. Apply Z-score normalization to Assignment_Score after imputing the missing value.

iii. Compare the transformed results.

C. Discretization

Divide Study_Hours into three categories:

- Low
- Medium
- High

Define suitable thresholds and assign each student to a category.

D. Interpretation

Explain how data preprocessing can improve the accuracy and interpretability of a student-performance prediction model.

SIMATS ENGINEERING

ASSESSMENT TOOL 3: SCENARIO-BASED SOLUTIONS

Course Code: CSA1610 | Course Name: Data Warehousing and Data Mining

Course Outcome Covered (CO2): Apply suitable Data Pre-processing techniques (Cleaning, Integration, Transformation, Reduction)

Total Marks: 50 | Weightage: 100%

Question 1: Data Cleaning and Normalization in Healthcare Analytics

Scenario Overview: A healthcare analytics company is predicting patient readmission based on hospital dataset containing missing values, inconsistent categorical entries, and extreme values.

Tasks & Detailed Solutions

A. Missing Value Treatment

- i. Identification of Missing Values:
  - Age: Missing at Patient P2
  - BMI: Missing at Patient P3
  - Annual_Visits: Missing at Patient P5
- ii. Imputation Calculations:

  • Age (Known: 25, 42, 110, 36, 50, 29, 45):
    - Mean = 337 / 7 = 48.14 years
    - Median = 42 years (Sorted: 25, 29, 36, 42, 45, 50, 110)

  • BMI (Known: 22.5, 24.8, 45.2, 29.5, 31.2, 21.8, 60.0):
    - Mean = 235 / 7 = 33.57 kg/m²
    - Median = 29.5 kg/m² (Sorted: 21.8, 22.5, 24.8, 29.5, 31.2, 45.2, 60.0)

  • Annual_Visits (Known: 3, 5, 7, 50, 12, 4, 40):
    - Mean = 121 / 7 = 17.29 visits
    - Median = 7 visits (Sorted: 3, 4, 5, 7, 12, 40, 50)
- iii. Method Comparison & Justification:

Median Imputation is significantly more appropriate for this dataset. The presence of extreme outliers (e.g., Age 110, BMI 60.0, Annual_Visits 50) heavily skews the mean upward, resulting in unnaturally high estimates. Median imputation provides a robust central value that is resilient to extreme values.

B. Outlier Detection

- i. Z-Score Calculations (Using complete observations: Mean & SD):

  • Age (Mean = 48.14, SD = 28.17):

  - P1 (25): Z = -0.82 | P3 (42): Z = -0.22 | P4 (110): Z = +2.20 (Outlier)

  - P5 (36): Z = -0.43 | P6 (50): Z = +0.07 | P7 (29): Z = -0.68 | P8 (45): Z = -0.11

  • Annual_Visits (Mean = 17.29, SD = 18.57):

  - P1 (3): Z = -0.77 | P2 (5): Z = -0.66 | P3 (7): Z = -0.55 | P4 (50): Z = +1.76 (Extreme)

  - P6 (12): Z = -0.28 | P7 (4): Z = -0.72 | P8 (40): Z = +1.22

- ii. Extreme Observation Records:

Patient P4 (Age 110, Visits 50) and Patient P8 (BMI 60.0, Visits 40) contain extreme observations.

- iii. Treatment Approach & Justification:

• Age 110 (P4): Record removal or capping at upper age threshold (e.g., 80) is justified, as 110 is likely a data entry error or non-representative anomaly.

• BMI 60.0 & Visits 40/50: Capping (Winsorization) at the 95th percentile is recommended to retain sample size while preventing extreme gradients in ML models.

C. Categorical Standardization

- i & ii. Inconsistent Gender Mapping:

Patient ID	Original Categorical Entry	Standardized Categorical Entry
P1	M	Male
P2	Male	Male
P3	F	Female
P4	Female	Female
P5	female	Female
P6	FEMALE	Female
P7	male	Male
P8	Female	Female

- iii. Importance for Machine Learning Models:

ML models require standardized categorical values during One-Hot or Label Encoding. Without standardization, casing/abbreviation variants (e.g., 'M', 'Male', 'male') are created as distinct binary features, unnecessarily inflating dimensional space and diluting model learning signals.

D. Min-Max Normalization of BMI

- i. Normalized Values [0, 1] Range (Min = 21.8, Max = 60.0, Range = 38.2):

Patient ID	Original BMI	Min-Max Normalized BMI
P1	22.5	0.0183
P2	24.8	0.0785
P3	29.5 (Imputed)	0.2016
P4	45.2	0.6126
P5	29.5	0.2016
P6	31.2	0.2461
P7	21.8 (Min)	0.0000
P8	60.0 (Max)	1.0000

- ii. Effect of Normalization on Performance:

Min-Max normalization scales feature ranges into standard uniform bounds [0, 1]. It prevents high-magnitude features from dominating distance-based algorithms (k-NN, SVM) or parameter weights in neural networks, ensuring balanced convergence during gradient descent.

Question 2: Covariance and Feature Selection in Banking Analytics

Scenario Overview: Developing a customer-risk prediction model for bank clients while resolving feature redundancy, high correlation, and computational overhead.

Tasks & Detailed Solutions

A. Covariance Analysis

- i. Covariance Calculation between Income (X) and Loan_Amount (Y):

• Mean Income ($\bar{X}$) = 577 / 8 = 72.125 (₹K)

• Mean Loan_Amount ($\bar{Y}$) = 1650 / 8 = 206.25 (₹K)

• $\Sigma(X_i - \bar{X})(Y_i - \bar{Y})$ = 25,068.75

• Sample Covariance Cov(X,Y) = 25,068.75 / (8 - 1) = +3,581.25

- ii. Relationship Type: Strongly Positive Covariance (+3,581.25).
- iii. Interpretation: Indicates that as customer income increases, the requested/approved loan amount scales up proportionally.

B. Feature Relationship Analysis

- i. Highly Correlated Feature Pairs:
  - • Income & Loan_Amount
    • Income & Savings
    • Loan_Amount & Monthly_EMI
- ii. Reason for Redundancy: Monthly_EMI is a direct mathematical derivative of Loan_Amount. Higher income directly correlates with increased savings capacity and loan eligibility, making these attributes convey redundant variance.
- iii. Three Features Recommended for Removal:
  - 1. Monthly_EMI
    2. Savings
    3. Loan_Amount (or Income depending on domain preference)

C. Feature Selection

- i. Optimal Feature Subset: {Income, Credit_Score, Age}
- ii. Justification: Credit_Score reflects historical repayment capability; Income measures current financial baseline; Age acts as a demographic stability proxy. Dropping Monthly_EMI and Savings removes high multicollinearity while retaining predictive accuracy for Risk.
- iii. Computational Cost Reduction: Reduces model parameters, speeds up training matrix operations O(d) reduction, prevents overfitting, and lowers memory overhead.

D. Standardization (Z-Score Normalization)

- i. Standardized Numerical Attributes (Income μ=72.125, σ=31.91; Credit_Score μ=721.25, σ=62.21):

Customer ID	Income (₹K)	Z(Income)	Credit Score	Z(Credit Score)
B1	35	-1.16	650	-1.15
B2	50	-0.69	680	-0.66
B3	70	-0.07	720	-0.02
B4	100	+0.87	780	+0.94
B5	42	-0.94	660	-0.98
B6	65	-0.22	710	-0.18
B7	130	+1.81	820	+1.59
B8	85	+0.40	760	+0.62

- ii. Necessity of Scaling:

Credit_Score ranges from 600 to 850, whereas Income ranges from 35 to 130. Without scaling, numerical algorithm distance calculations would be disproportionately dominated by Credit_Score due to raw numerical magnitude.

Question 3: Discretization of Sales Data in Retail Analytics

Scenario Overview: A retail company discretizes continuous Monthly_Spend data into intuitive categorical tiers for strategic executive decision-making.

Tasks & Detailed Solutions

A. Equal-Width Binning (k = 4 Bins)

- i & ii. Bin Width Calculation:
  • Bin Width = (Max - Min) / k = (6500 - 500) / 4 = ₹1,500
- iii. Interval Definition & Customer Mapping:

• Bin 1 [₹500, ₹2,000): C1 (500), C2 (750), C3 (1000), C4 (1500)

• Bin 2 [₹2,000, ₹3,500): C5 (2000), C6 (2500), C7 (3000)

• Bin 3 [₹3,500, ₹5,000): C8 (4000)

• Bin 4 [₹5,000, ₹6,500]: C9 (5000), C10 (6500)

B. Equal-Frequency Binning (4 Bins)

- i & ii. Customer Bin Assignments (Approx 2-3 records per bin):

• Bin 1 (2 records): C1 (500), C2 (750)

• Bin 2 (3 records): C3 (1000), C4 (1500), C5 (2000)

• Bin 3 (2 records): C6 (2500), C7 (3000)

• Bin 4 (3 records): C8 (4000), C9 (5000), C10 (6500)

- iii. Comparison:

  • Equal-Width creates uniform range sizes but yields skewed customer counts per bin.

  • Equal-Frequency balances sample count across bins, preserving percentile representation.

C. Concept Hierarchy Construction

Monthly_Spend → Categorical Hierarchy:

• Low: ₹500 – ₹1,200 (Customers C1, C2, C3)

• Medium: ₹1,201 – ₹2,800 (Customers C4, C5, C6)

• High: ₹2,801 – ₹4,500 (Customers C7, C8)

• Very High: > ₹4,500 (Customers C9, C10)

D. Interpretation

- i. Decision-Making Benefits: Enables business managers to execute targeted loyalty marketing per spend tier rather than interpreting noisy raw continuous numbers.
- ii. Disadvantage: Loss of granular intra-bin continuous information (e.g., treating ₹500 and ₹1,500 identically in the same broad category).

Question 4: Data Transformation in Manufacturing Quality Analytics

Scenario Overview: Transforming machine operating temperatures and evaluating outlier sensitivity for quality assurance analytics.

Tasks & Detailed Solutions

A & B. Min-Max and Z-Score Transformation Results

- Stats: Min = 55°C, Max = 120°C, Range = 65°C | Mean μ = 73.375°C, Standard Deviation σ = 20.35°C

Machine ID	Temperature (°C)	Min-Max Normalized [0,1]	Z-Score	Outlier Flag
M1	55	0.0000	-0.90	Normal
M2	60	0.0769	-0.66	Normal
M3	62	0.1077	-0.56	Normal
M4	65	0.1538	-0.41	Normal
M5	70	0.2308	-0.17	Normal
M6	75	0.3077	+0.08	Normal
M7	80	0.3846	+0.33	Normal
M8	120	1.0000	+2.29	Extreme Outlier

- Observation with Maximum Normalized Value: Machine M8 obtains 1.0000.

C. Outlier Analysis

- i. Evaluation: 120°C yields a Z-score of +2.29 (> 2.0 SD limit), confirming it as an extreme statistical outlier relative to operational baseline (55–80°C).
- ii. Effect on Min-Max Normalization: Compresses all standard operating temperatures (55–80°C) into a narrow normalized range (0.00 to 0.38), reducing sensitivity across normal operating ranges.
- iii. Treatment Suggestion: Winsorization (capping upper bound at 85°C) or isolating the anomaly in a separate alert log.

D. Sensitivity Comparison

• Min-Max Normalization: Highly sensitive to extreme outliers; a single extreme value drastically skews the scale and compresses all other values.
• Z-Score Normalization: Moderately sensitive (outliers affect mean and SD), but avoids hard bounding and retains standard distribution proportions.

Question 5: Missing Value Imputation & Transformation in Student Analytics

Scenario Overview: Imputing missing academic performance indicators, standardizing test metrics, and discretizing study habits.

Tasks & Detailed Solutions

A. Missing Value Analysis

- i. Identified Missing Entries:

  • Internal_Mark missing at S2

  • Attendance missing at S3

  • Study_Hours missing at S5

  • Assignment_Score missing at S6

- ii. Imputation Values:

  • Study_Hours: Mean = 5.86, Median = 5

  • Internal_Mark: Mean = 79.43, Median = 80

  • Attendance: Mean = 86.43, Median = 90

  • Assignment_Score: Mean = 76.00, Median = 78

- iii. Suitable Method for Study_Hours: Median Imputation (5 hours) is more appropriate because high study hours (S8 = 12) skew the mean upward.

B. Data Transformation & Comparison

Student ID	Internal_Mark	Min-Max (Internal)	Assignment_Score	Z-Score (Assignment)
S1	55	0.000	60	-1.35
S2	80 (Imputed)	0.581	65	-0.93
S3	68	0.302	72	-0.34
S4	75	0.465	78	+0.17
S5	80	0.581	82	+0.51
S6	88	0.767	76 (Imputed)	0.00
S7	92	0.860	90	+1.18
S8	98	1.000	95	+1.61

C. Discretization of Study_Hours

- Defined Thresholds:

  • Low: < 4 hours (S1, S2)

  • Medium: 4 to 7 hours (S3, S4, S5-Imputed, S6)

  • High: > 7 hours (S7, S8)

D. Interpretation

Preprocessing missing values prevents sample loss; feature scaling ensures fair evaluation across attributes of varying ranges; discretization converts numerical study hours into operational academic risk categories for targeted student interventions.